

In this assignment, we fine-tuned BERT on NER tasks. The data used to train this model included train, dev, and test tsv's that contained BIO-formatted data. The token and label lists from the files were wrapped into Hugging Face Dataset objects in preparation for model training. Further data preparation was needed due to BERT normally using subword tokenization. BIO labels were assigned to the first subword piece, and special tokens and remaining subword pieces were assigned with -100. For the test set, a value of 0 was assigned to the first subword piece since the test set has no ner_tags.

Moving onto the model, a pretrained bert-base-cased model was loaded and had a token-classification head attached. To evaluate the model, precision, recall, F1 score, and accuracy were measured, and the optimal model was chosen according to the F1 score. In training the model, a standard set of hyperparameters for this type of task were initially chosen. Using the dev set, the learning rate and number of epochs hyperparameters were tuned independently. With the highest independent F1 scores when altering learning rate and number of epochs, a final tuning was done using the learning rate and number of epochs that gave the best F1 score. This final model performed the best out of all the models trained, with an F1 score of 0.9122. The selected model then generated predictions on the test set, outputting test_predictions.tsv. The F1 score of the selected model using the test set was 0.8965.

For error analysis, I wanted to look deeper into the metrics returned by the model. To do this, I used the dev set to measure the precision for each BIO label. I found that for O, the model performed very well with a precision of 0.995. For B, the model had a precision of 0.926, and for I, the model had a precision of 0.837. Based on these results, I believe the model does very well at identifying non-chemical entities (O), but performs slightly worse with chemical entities, particularly when the chemical entity is composed of multiple words.